

OSTF Paper Solution

Q.1 MCQs (1 mark × 10 = 10 Marks)

1. Two main branches of Statistics → **Descriptive and Inferential Statistics** ✓ (1 Mark)
 2. Dependent variable in Pima Indians Diabetes Dataset → **Outcome (0 or 1)** ✓ (1 Mark)
 3. Measures of centrality → **Mean, Median, Mode** → **All** ✓ (1 Mark)
 4. Sampling method dividing into groups and selecting entire groups → **Cluster Sampling** ✓ (1 Mark)
 5. Types of hypothesis tests → **Z-test, ANOVA, Chi-square** → **All** ✓ (1 Mark)
 6. P-value indicates → **Statistical significance** ✓ (1 Mark)
 7. Test used to compare means of two different samples → **Two-sample t-test** ✓ (1 Mark)
 8. Foundational rules of probability → **All: P(Impossible)=0, P(Certain)=1, Addition rule** ✓ (1 Mark)
 9. Degrees of freedom (DF) → **N-1** ✓ (1 Mark)
 10. Conditional probability → **P(A|B)** ✓ (1 Mark)
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Q.2 Error Finding (2 Marks each × 4 = 8 Marks)

Q1. Error Finding and Correction

1. Printing list elements

Error Code:

```
numbers = [1, 2, 3, 4, 5]
for i in range(0, len(numbers)):
    print(numbers[i])
print("Done"
```

Error:

- Missing closing parenthesis) in print("Done").

Correct Code:

```
numbers = [1, 2, 3, 4, 5]
for i in range(0, len(numbers)):
    print(numbers[i])
print("Done")
```

2. Function call

Error Code:

```
def add(a, b):
    return a + b
```

```
result = add(5)
print(result)
```

Error:

- Function add requires **two arguments**, but only one was given.

Correct Code:

```
def add(a, b):
```

```
return a + b
```

```
result = add(5, 3)
```

```
print(result)
```

3. List index

Error Code:

```
my_list = [10, 20, 30]
```

```
print(my_list[3])
```

Error:

- Index 3 is **out of range** (valid indexes are 0, 1, 2).

Correct Code:

```
my_list = [10, 20, 30]
```

```
print(my_list[2]) # prints 30
```

4. Import statement

Error Code:

```
import numppy as np
```

```
arr = np.array([1, 2, 3])
```

```
print(arr)
```

Error:

- Typo in module name → should be numpy, not numppy.

Correct Code:

```
import numpy as np
```

```
arr = np.array([1, 2, 3])
```

```
print(arr)
```

5. Variable naming

Error Code:

```
2value = 10  
print(2value)
```

Error:

- Variable names **cannot start with a number**.

Correct Code:

```
value2 = 10  
print(value2)
```

Q.3 Attempt Any TWO (5 Marks each = 10 Marks)**(i) Data Preprocessing Steps (5 Marks)**

- **Data Cleaning** → Handle missing values (drop, fill with mean/median).
 - **Data Transformation** → Scaling/normalization for uniformity.
 - **Data Encoding** → Convert categorical data (Label Encoding, One-Hot Encoding).
 - **Outlier Detection** → Remove abnormal data points.
 - **Feature Selection** → Keep only useful variables.
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(ii) Perception and Actuation in Data Science (5 Marks)

- **Perception** → Gathering and interpreting data from environment (e.g., sensors, user inputs). In Data Science, it means collecting data from sources (databases, IoT devices, web scraping).
 - **Actuation** → Taking actions/decisions based on processed data (e.g., sending alert, recommendation, controlling a device).
 - Example: In a smart home system, perception = detecting temperature, actuation = turning on AC.
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(iii) Hypothesis in Data Science (5 Marks)

- A **hypothesis** is an assumption tested using data.
 - In Data Science:
 - Helps validate models.
 - Used in A/B testing for product improvement.
 - Example: *"New website design increases user engagement"*.
 - **Hypothesis testing** ensures data-driven decision-making in engineering/business.
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Q.4 Programs (5 Marks each = 10 Marks)

(a) Factorial Program (5 Marks)

```
num = int(input("Enter a number: "))
fact = 1
for i in range(1, num+1):
    fact *= i
print("Factorial of", num, "is", fact)
```

(b) Even or Odd Program (5 Marks)

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print(num, "is Even")
else:
    print(num, "is Odd")
```

OR (b) Grade Sheet Program (5 Marks)

```
marks = int(input("Enter student marks: "))

if marks >= 90:
    grade = "A"
```

```
elif marks >= 75:
```

```
    grade = "B"
```

```
elif marks >= 60:
```

```
    grade = "C"
```

```
elif marks >= 40:
```

```
    grade = "D"
```

```
else:
```

```
    grade = "F"
```

```
print("Marks:", marks, "Grade:", grade)
```